JING-TONG TZENG

Hsinchu, Taiwan   roger37890426@gmail.com   (+886)978-472-577

EDUCATION

National Tsing Hua University, Hsinchu, Taiwan *Sep. 2017 - Jan. 2022*
Bachelor of Power Mechanical Engineering (Minor in Electrical Engineering)
- Overall GPA:3.71/4.3

National Tsing Hua University, Hsinchu, Taiwan *Feb. 2022 - Nov. 2024*
Master of College of Semiconductor Research
- Overall GPA:4.01/4.3
- **Research field**: Multitask Learning, Multimodal Learning, Generative Model, Speech and Language Processing

WORK EXPERIENCE

Behavioral Informatics & Interaction Computation Lab **National Tsing Hua University, Taiwan**
Research Assistant *Nov. 2024 - Aug.2025*

- Collaborated on industry-academic joint research projects, applying machine learning techniques to real-world challenges in signal processing and healthcare applications.
- Mentored graduate students, providing guidance on research methodologies, deep learning architectures, and experimental design.

Multimodal Signal Processing Lab **The University of Texas at Dallas, USA**
Visiting Researcher *Apr. 2024 - Sep. 2024*

- Proposed a shared self-supervised speech representation framework integrating speech enhancement and speech emotion recognition, achieving a 1.8% improvement in F1-Macro score without compromising SE performance.

RESEARCH PROJECT

Behavioral Informatics & Interaction Computation Lab **National Tsing Hua University, Taiwan**
Advisor: Prof. Chi-Chun (Jeremy) Lee

Topic: Noise Robust Speech Emotion Recognition (with CMU MSP Lab, Pittsburgh) *Nov. 2024 - Sep. 2025*

- Introduced a novel application of Mixture of Experts within WavLM layers to effectively balance speech enhancement and speech emotion recognition, achieving a 3.4% improvement in F1-macro score.

Topic: Multimodal Speech Emotion Recognition *Nov. 2024 - Feb. 2025*

- Employed a multimodal fusion strategy to integrate textual and acoustic features, achieving the highest valence score in the Interspeech 2025 Speech Emotion Recognition in Naturalistic Conditions Challenge.

Topic: Respiratory Sound Classification System *Feb. 2022 - Nov. 2024*

- Proposed a novel noise robust respiratory sound classification system leveraging speech enhancement techniques, resulting in a notable 20.2% increase in sensitivity and a substantial 21.9% improvement in specificity.
- Developed a patch-wise gamma correction augmentation that better captures continuous and discontinuous spectrogram patterns, improving sensitivity by 11.8% over state-of-the-art methods.

Topic: Heart Murmur Classification System *Sep. 2023 - Sep. 2024*

- Applied masked autoencoders to reconstruct missing cardiac recordings, reducing mean outcome-classification cost by 44.12% and outperforming prior state-of-the-art imputation methods.

Topic: Automatic Kidney Cancer Grading System *Sep. 2023 - Sep. 2024*

- Co-developed a segmentation mask augmentation approach, improving AUC by an average of 5.8% across diverse segmentation errors, model architectures, and datasets versus conventional augmentation.

- Utilized nnUNet for early-stage tumor segmentation, achieving impressive Dice scores of 96.16% for kidneys, 84.79% for tumors, and 71.49% for cysts.

HONORS AND AWARDS

IEEE ICASSP 2025 Conference Travel Grant	*IEEE Signal Processing Society*
Student International Visiting Scholarship Award 2024	*National Tsing Hua University*
Awarded College Student Research Fellowship	*Ministry of Science and Technology (MOST)*
Japan Formula SAE 2020 9th Place	*Society of Automotive Engineers*

TECHNICAL SKILLS

- **Programming Languages**: C/C++, Python, MATLAB, Shell, Verilog
- **Technologies**: PyTorch, AWS, Docker, Linux, Git

PUBLICATION

[1] **Tzeng, J. T.**, Busso, C., & Lee, C. C. **Joint learning using mixture-of-expert-based representation for enhanced speech generation and robust emotion recognition,** submitted to TASLP (under review).

[2] Chen, X. Y., **Tzeng, J. T.**, Busso, C., & Lee, C. C. **Formant-Guided Speech Repair for Enhanced Comprehension of Mandarin Dysarthric Speech,** submitted to ICASSP 2026 (under review).

[3] Yen, Y. W., **Tzeng, J. T.**, Chang, A. Y., Huang, C. H., Huang, E. P. C., & Lee, C. C. **Multiple Instance Learning for Temporal Localization of Abnormal Respiratory Sounds,** submitted to ICASSP 2026 (under review).

[4] Yang, J. S., **Tzeng, J. T.**, & Lee, C. C. (2025). **Personalized Federated Learning with Fuzzy Clustering for Dysarthric Speech Recognition,** in ASRU 2025-2025 IEEE Automatic Speech Recognition and Understanding Workshop.

[5] **Tzeng, J. T.**, Su, B. H., Wu, Y. T., Chou, H. H., & Lee, C. C. (2025). **Lessons Learnt: Revisiting Key Training Strategies for Effective Speech Emotion Recognition in the Wild,** in Proceedings of Interspeech 2025.

[6] **Tzeng, J. T.**, Li, J. L., Chen, H. Y., Huang, C. H., Chen, C. H., Fan, C. Y., Huang, E. P. C., & Lee, C. C. (2025). **Improving the Robustness and Clinical Applicability of Automatic Respiratory Sound Classification Using Deep Learning–Based Audio Enhancement: Algorithm Development and Validation,** in Journal of Medical Internet Research AI.

[7] **Tzeng, J. T.**, Leem, S. G., Salman, A., Lee, C. C. & Busso, C. (2025, April). **Noise-Robust Speech Emotion Recognition Using Shared Self-Supervised Representations with Integrated Speech Enhancement,** in ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing.

[8] Chang, A. Y., **Tzeng, J. T.**, Chen, H. Y., Huang, C. H., Huang, E. P. C., & Lee, C. C. (2025, April). **Valve Token Masked Autoencoder for Missing Recordings on Cardiac Abnormality Classification,** in ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing.

[9] Weng, C. C., Chen, H. Y., **Tzeng, J. T.**, Lin, C. H., Kuo, P. C., & Lee, C. C. (2025, April). **Mask Augmentation For Tumor Classification In Medical Images,** in ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing.

[10] Huang, C. H., Chen, C. H., **Tzeng, J. T.**, Chang, A. Y., Fan, C. Y., Sung, C. W., ... & Huang, E. P. C. (2024). **The unreliability of crackles: insights from a breath sound study using physicians and artificial intelligence,** in NPJ Primary Care Respiratory Medicine, 34(1), 28.

[11] Chang, A. Y., **Tzeng, J. T.**, Chen, H. Y., Sung, C. W., Huang, C. H., Huang, E. P. C., & Lee, C. C. (2024, April). **GaP-Aug: Gamma Patch-Wise Correction Augmentation Method for Respiratory Sound Classification,** in ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing.

ACADEMIC SERVICE

- Reviewer, *IEEE/ACM Transactions on Audio, Speech, and Language Processing*
- Reviewer, *APSIPA Transactions on Signal and Information Processing*